

DebateGNN: A Multitask Graph Neural Network for Echo Chamber Detection in Online Debates

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Abstract—Echo chambers (ECs) are often measured through endorsement proxies such as retweets or likes, which capture homophily but potentially overlook argumentative dynamics. This thesis examines whether incorporating stance information into message passing alters how ECs are quantified in debate settings, using Reddit reply graphs where stance signals (agree, neutral, disagree) shape community structure. We introduce DebateGNN, a hybrid Graph Neural Network that combines GraphSAGE and Edge-Conditioned Convolutions. Edges encode annotator-derived stance vectors and confidence scores, and the model is trained with a multitask objective for link, confidence and stance predictions. Using the DEBAGREEMENT dataset of stance-labelled Reddit replies, we compute the Echo Chamber Score (ECS) over temporal graph snapshots and compare DebateGNN embeddings to those from the GCN-based EchoGAE baseline. Results show that EchoGAE generally produces higher ECS values and closer alignment with structural partitions, while DebateGNN yields typically lower trajectories and performs better in low-modularity, disagreement-heavy settings. We conclude that ECS outcomes depend on modelling choices, and recommend reporting both topology-led and stance-aware estimates. Code and pipelines are available on GitHub.

Index Terms—Graph Neural Networks, Echo Chambers, Online Social Networks, Stance Detection, Temporal Graphs, Community Detection

I. INTRODUCTION

Online debates on social networks have become a central arena for public discourse across politics, culture and science. A recurring pattern within these debates is the emergence of *echo chambers* (ECs), which are characterised by clusters of users whose interactions are internally reinforcing and externally sparse (Cinelli et al. 2021). Conventionally, ECs have often been explained through *homophily*, the tendency for individuals to connect with like-minded others (McPherson, Smith-Lovin, and Cook 2001). Yet in debate contexts, interactions frequently involve disagreement or conflict; even antagonistic exchanges can reinforce group boundaries and polarisation dynamics (Bail et al. 2018). More broadly, EC formation reflects multiple interaction forces: the threaded and asymmetric nature of reply discussions, which organises debates into rooted trees of directed exchanges (Medvedev, Lambiotte, and Delvenne 2019; Yu, Jiang, and Dhillon 2024; Goglia and Vega 2024), as well as platform-level mechanisms such as algorithmic content ranking and self-selection into topical communities (e.g., subreddits) (Bakshy, Messing, and Adamic 2015; Flaxman, Goel, and Rao 2016). Taken together,

these processes vary across platforms but generally amplify within-cluster contact while suppressing cross-cutting interactions (Cinelli et al. 2021).

Despite their prevalence, however, ECs and their measurement in debate settings remain incompletely understood (Mahmoudi, Jemielniak, and Ciechanowski 2024; Hartmann et al. 2025). Conceptually, many studies rely on endorsement proxies such as retweets or likes, which capture homophily but elide the direction and intensity of stance signals (e.g. negative ties) that better represent polarised debates (Metaxas et al. 2021; Keuchenius, Törnberg, and Uitermark 2021). Empirically, large-scale analyses also concentrate on Twitter-like platforms, while Reddit’s long-form, reply-threaded debates remain comparatively under-examined (Hartmann et al. 2025). Yet recent work highlights Reddit’s unique dynamics, including nuanced ideological alignments and evolving hostilities over time (Efstratiou et al. 2023). Methodologically, computational approaches to ECs also remain fragmented. Existing models typically capture only one or two functional dimensions: (i) *structural insularity* (preferential interaction within like-minded groups), (ii) *semantic alignment* (ideological or linguistic similarity) or (iii) *reinforcement dynamics* (belief amplification over time), with less than 30% addressing reinforcement dynamics and fewer than 50% examining structural insularity (Mahmoudi, Jemielniak, and Ciechanowski 2024). In debate contexts, where structure, semantics and temporal dynamics are deeply intertwined, this fragmentation significantly limits systematic characterisation.

There is also no consensus on the net effects of ECs. Recent reviews converge on the above three functional dimensions (Mahmoudi, Jemielniak, and Ciechanowski 2024; Hartmann et al. 2025), while highlighting heterogeneous operationalisations and outcomes. Importantly, just as disagreement in debates is not inherently harmful (and may be necessary in fostering deliberation and perspective-taking), there is no evidence that ECs exert solely negative impacts. While many works associate them with polarisation and misinformation, others document cases where dense, aligned communities enable efficient coordination or accelerate rumour rebuttal during crises (Mahmoudi, Jemielniak, and Ciechanowski 2024; D. Wang et al. 2022).

Despite definitional fragmentation, computational studies show that ECs can be measured. For instance, the Echo

Chamber Score (ECS) (Alatawi, Sheth, and Liu 2024) moves beyond bipartisan framings by quantifying EC strength as a distance-based measure of within-community cohesion and across-community separation in latent space. However, ECS was validated mainly on retweet networks, where links are treated as endorsements. In debates, where interactions often encode disagreement, this assumption may not hold.

Addressing these challenges requires models that integrate structural, semantic and dynamic factors. *Graph Neural Networks* (GNNs) are widely used in social network analysis for their ability to jointly learn from node features and relational structure through message passing; their expressive capacity has proven effective across tasks such as user classification (Zhou et al. 2020; Z. Wu et al. 2021), influence modelling (Qiu et al. 2018) and community detection (Kojaku et al. 2024), making them a promising foundation for studying ECs in debate settings.

As measures like ECS indicate, any quantification of ECs depends on how underlying communities are defined within a network. Fundamentally, communities are characterised structurally, as cohesive groups of nodes with higher within-group edge density (Traag, Waltman, and Eck 2019; Kojaku et al. 2024). In our work, we treat an EC as a special case of a community where structural cohesion is coupled with semantic alignment (e.g., shared stance or opinion). We therefore assume that while the core idea (behind community detection) is shared, the nuance and operationalisation differ across user-declared groups, connectivity-based partitions (e.g., Louvain, Leiden) and embedding-based partitions (e.g., GNNs). Even operationally, these definitions may not align, as connectivity methods assume cohesive subgraphs, whereas embedding methods assume geometric separability in a latent space (Traag, Waltman, and Eck 2019; Kojaku et al. 2024).

This thesis proposes **DebateGNN**, a stance-aware multitask GNN designed for online debate graphs. The model integrates scalability with expressivity by hybridising GraphSAGE (Hamilton, Ying, and Leskovec 2017) and Edge-Conditioned Convolution (ECC) (Simonovsky and Komodakis 2017) layers. GraphSAGE layers provide inductive, multi-hop relation encoding, while ECC conditions message passing directly on edge attributes: annotator stance labels (agree, neutral, disagree) and confidence scores derived from annotation quality metrics. A joint learning objective couples link prediction, stance distribution estimation and confidence regression within an uncertainty-weighted multitask framework. This design enables DebateGNN to move beyond homophily-based assumptions by directly embedding the signals that mediate stance alignment in debates. We build on the **DEBAGREEMENT** dataset (Pougué-Biyong et al. 2021), a Reddit comment-reply dataset containing fine-grained stance annotations and temporal interaction data from five socio-political subreddits.

Paper Contributions. Key contributions of this thesis include:

- **Stance-Aware Representation:** Introduce ternary stance vectors (agree, neutral, disagree) with annotator-derived

confidence scores as edge attributes, enabling richer relational modelling than prior retweet- or like-based graphs.

- **DebateGNN Framework:** Develop a hybrid multitask GNN combining GraphSAGE and Edge-Conditioned Convolutions, propagating stance- and confidence-conditioned signals with a multitask decoder for link, confidence, and stance prediction.
- **Temporal ECS Analysis:** Extend the Echo Chamber Score (ECS) to temporal debate graphs using GNN embeddings and community detection, demonstrated through a detailed `r/brexit` case study of community dynamics.
- **Open-Source Implementation:** Release all code, pre-processing pipelines and evaluation scripts on [GitHub](#) to support replication and future research.

II. RELATED WORK

A. Echo Chamber Foundation and Taxonomy

In their systematic literature review of 28 studies published between 2013 and 2022, Mahmoudi, Jemielniak, and Ciechanowski (2024) identify three key dimensions of EC formation: (i) *structural insularity*, (ii) *semantic alignment* and (iii) *reinforcement dynamics*. *Structural insularity* reflects the topological isolation of like-minded groups, often measured using modularity, clustering coefficients or community detection. *Semantic alignment* captures shared beliefs, quantified through stance polarity, embedding similarity or purity metrics (the proportion of ideologically similar members). *Reinforcement dynamics*—the gradual strengthening of group consensus over time—has been less explored, and is typically inferred from temporal shifts in stance distributions or declining content/exposure diversity in recommender systems. Despite the recent expansion of computational EC research, there has yet to be a modelling approach that jointly integrates all three functional dimensions (Mahmoudi, Jemielniak, and Ciechanowski 2024).

This section discusses prior approaches to EC modelling, which can be grouped into two broad paradigms: (i) *non-deep learning* and (ii) *deep learning*.

B. Non-Deep Learning Approaches

1) *Statistical Network Models:* Statistical models infer ECs directly from structural interaction patterns, often without direct use of textual stance data. For instance, D. Wang et al. (2022) construct co-retweet and co-comment networks on Weibo to examine user interactions during COVID-19 rumour rebuttals. In their formulation, undirected, weighted edges connect posts co-engaged by users, yielding a shared-audience interaction graph. Tie formation is then modelled using Exponential Random Graph Models (ERGMs) with nodematch terms, estimating the likelihood that proximate nodes share attributes (e.g., stance or topic) to capture homophily-driven clustering in polarised discourse. Building on a similar structural premise, Perera and Karunasekera (2024) propose *SEcho*, a signed-graph framework that explicitly accounts for both agreement and disagreement. Their approach applies a greedy

optimisation over *signed modularity* (an extension of modularity that rewards dense positive ties within communities while penalising negative ties across them) to identify clusters with strong internal cohesion and systematic external opposition.

Together, these approaches demonstrate how structural models can reveal patterns of insularity and polarisation. However, they remain semantically agnostic—treating opinions as abstract node attributes or signed edges rather than grounded linguistic positions—and therefore overlook belief homogeneity and reinforcement dynamics.

2) *Agent-Based Opinion Dynamics Models*: Agent-Based Models (ABMs) simulate how agents update their opinions under local interaction rules, enabling controlled experiments on factors shaping EC formation (e.g., homophily, clustering, activity levels). Unlike statistical models, which infer patterns from observed structure, ABMs generate synthetic dynamics and capture micro-to-macro feedback loops that drive clustering and polarisation. Baumann et al. (2020) model EC formation in temporal social networks through homophily-biased interactions: at each timestep, agents form directed ties with probability inversely proportional to their opinion distance, $p_{ij} \propto |x_i - x_j|^{-\beta}$. With power-law activity distributions, highly active users disproportionately reinforce like-minded links, producing emergent ECs. Building on this framework, Botte, Ryckebusch, and Rocha (2022) analyse the effects of local clustering and user stubbornness. They show that EC size grows with higher clustering but contracts when agents exhibit greater stubbornness, while global modularity has comparatively little impact. Similarly, Pranesh and Gupta (2025) extend Baumann et al.’s model by incorporating cognitive inertia, i.e., the resistance to opinion change. Through steady-state analysis, they demonstrate that persistent, polarised clusters arise under high inertia, with ECs manifesting as bimodal opinion distributions.

Overall, ABMs offer insight into polarisation mechanisms in controlled experiments, but share the same semantic agnosticism and thus limitations as statistical network models.

C. Deep Learning Approaches

1) *NLP-Only Models*: Content-centric approaches rely on NLP techniques to infer semantic alignment from text, treating language as the primary signal of ideological homogeneity while largely abstracting away network structure. Early work by Calderón et al. (2019) introduced *ECHO*, a stance- and emotion-based classifier that labels Facebook post-comment pairs as echoing or non-echoing. Their Echo Chamber Index (ECI), defined as the proportion of echoing responses within a thread, captures local sentiment alignment but remains limited to the micro-level, without modelling user relations or community organisation. Rosen and Dale (2023) extended this to group-level convergence, using BERT embeddings to quantify discourse entropy across Reddit communities. They show intra-group language becomes increasingly predictable over time, particularly when conditioned on contextual variables such as upvotes or author identity. However, this framework still omits reply-graph structure and interaction topology. X.

Wang, J. Li, and Rajtmajer (2024) move closer to integrating content and structure by identifying ECs as strongly connected components (SCCs) in Twitter reply networks, then analysing linguistic features within them (e.g., collective pronouns, syntactic simplification, causal connectives). While highlighting shared narrative strategies in misinformation clusters, their model does not incorporate edge-level stance or temporal evolution.

Overall, NLP-based approaches reveal fine-grained linguistic indicators of alignment and convergence, but their abstraction from interaction structure limits their capacity to capture how semantics and topology jointly drive EC formation.

2) *Hybrid Text-Graph Approaches*: Recent models integrate structural and semantic signals to improve echo chamber detection, aiming to capture both topological cohesion and ideological alignment. These generally fall into two categories: (i) pipelines that combine NLP-based stance classification with graph analysis and (ii) GNN-based hybrids that learn joint representations end-to-end.

a) *Pipeline-based Integration*: Cau, Morini, and Rossetti (2024) propose a platform-independent pipeline for detecting and tracking ECs across time. User stances are classified with BERT and embedded into temporal interaction graphs from Reddit and Twitter/X debates. Communities are identified via EVA (Evidential Label Propagation), which optimises modularity and semantic purity, and temporal stability is then measured by tracking user retention across snapshots with Jaccard similarity. Although effective at capturing persistence, the model treats content as static opinion labels and does not model semantic alignment within or across communities.

b) *GNN-based Hybrids*: While pipeline methods sequentially combine stance classification with graph clustering, GNN-based hybrids integrate structure and content directly by learning joint representations. Minici et al. (2022) propose the Echo Chamber Detection (ECD) model: a probabilistic generative framework in which latent communities jointly explain both social links and information cascades. A two-layer Graph Convolutional Network (GCN) processes the social graph to estimate user-level latent variables, *polarised engagement* $\theta_{c,u}$ and *social engagement* $\phi_{c,u}$, which feed the probabilistic component. Each community is assigned a polarity score η_c , and the likelihood of observing a cascade D_i with polarity p_i depends on ideological concordance (i.e., $\text{sign}(p_i) = \text{sign}(\eta_c)$) weighted by user engagement. ECD outperforms a standalone GCN baseline on stance detection across synthetic and real-world datasets (Brexit, VaxNoVax, Referendum) but does not incorporate temporal reinforcement dynamics.

A parallel line of work replaces the generative component with self-supervised graph autoencoding: EchoGAE (Alatawi, Sheth, and Liu 2024) learns ideology-aware embeddings by combining user tweet text (Sentence-BERT) with the social adjacency A . A two-layer GCN fuses these signals into node embeddings Z , from which the *ECS* is computed to quantify ideological cohesion and separation. For a user u in community ω , cohesion λ_u is the average intra-community

embedding distance, and separation Δ_u is the distance to the nearest external community. The community-level ECS is:

$$\text{ECS}^*(\omega) = \frac{1}{|\omega|} \sum_{u \in \omega} \frac{\Delta_u - \lambda_u}{\max(\Delta_u, \lambda_u)}, \quad (1)$$

and the overall ECS is the average across communities (higher indicates greater insularity):

$$\text{ECS}(\Omega) = \frac{1}{|\Omega|} \sum_{\omega \in \Omega} \text{ECS}^*(\omega). \quad (2)$$

ECS depends only on an embedding space and a community partition, making it model-agnostic and imposing no assumptions on interaction-graph structure (Alatawi, Sheth, and Liu 2024). However, prior evaluations (EchoGAE) use static, retweet-based graphs with a GCN encoder, treating retweets as endorsements and omitting explicit edge stance and temporal reinforcement.

In our study, we retain ECS but change the representation by incorporating stance-aware, temporally segmented Reddit reply graphs with embeddings learned by a hybrid inductive GNN with an edge-conditioned final layer (see Sec. III-D). This setup lets us assess ECS’s portability and the value of explicit edge stance under temporal evaluation.

III. METHODOLOGY

A. Dataset

We use the **DEBAGREEMENT** dataset (Pougué-Biyong et al. 2021), a large-scale stance-labelled Reddit corpus comprising 42,894 annotated comment-reply pairs across five socio-political subreddits: r/BlackLivesMatter, r/brexit, r/climate, r/democrats and r/Republican. For r/brexit and r/climate, the dataset spans from the subreddit’s inception, whereas the remaining subreddits span from January 2021 to May 2021. Each comment-reply pair is labelled via a ternary stance scheme (“agree”, “neutral”, “disagree”). Labels are crowdsourced from 3–5 annotators using Scale AI’s dynamic consensus model, with additional quality metrics including the agreement fraction (proportion of annotator consensus) and Fleiss’ κ (inter-annotator reliability). Roughly one-third of examples achieved full annotator agreement, and expert adjudication was used to resolve tie cases. Importantly, interactions represent depth-1 reply chains within their subreddits rather than full discussion threads. Each child comment (reply) responds directly to a parent comment, which in turn responds to a submission post. Deeper nested replies were removed under the assumption that responses to top-level comments more reliably reflect durable ideological positions, whereas deeper chains may capture transient conversational dynamics (Pougué-Biyong et al. 2021).

B. Data Preparation

1) *Data Cleaning and Normalisation*: Subreddit names and stance labels were normalised and mapped to integer identifiers, while column names were harmonised for graph construction (e.g., aligning author and timestamp fields). All timestamps were converted to a consistent format. Self-replies,

which accounted for approximately 1% of interactions, were removed as they do not constitute meaningful social exchanges. This filtering resulted in a final dataset of 42,770 reply-pairs.

2) *Temporal Segmentation*: To enable longitudinal analysis, comments were segmented into fixed-length windows. Because the raw dataset provided timestamps only for child comments (replies), parent comment times were inferred by shifting them 30 minutes earlier to preserve causal ordering. We chose subreddit-specific window sizes (ranging from 30 to 90 days) to prevent graphs from becoming overly sparse or overly dense. To further balance interaction density, we iteratively merge windows with fewer than 100 comments into adjacent intervals (with preference for the smaller neighbour), after which timesteps were reindexed and effective window sizes and intervals were recomputed. Each comment (and by extension, each reply) was thereby assigned a timestep index, interval label and an observed window span in days. Table I summarises subreddit-level reply distributions, stance proportions and temporal segmentation characteristics.

3) *Confidence Computation*: For each reply-pair, we computed a confidence score w_i that reflects both annotator agreement and inter-annotator reliability:

$$w_i = \begin{cases} \hat{p}_i \cdot \kappa_i, & \text{if Fleiss' } \kappa_i \text{ is available} \\ 0.05, & \text{if } \kappa_i \text{ is missing} \end{cases} \quad (3)$$

where $\hat{p}_i$ is the fraction of annotators selecting the majority stance, and κ_i is the corresponding Fleiss’ κ . A fallback confidence level of 0.05 was used to indicate minimal reliability while retaining the interaction (approximately 80 cases).

4) *Aggregation of User Interactions*: We aggregated reply-pairs into directed user-user interactions within each subreddit and timestep. For every source–destination (i.e. child–parent) pair, multiple replies were collapsed into a single edge representation. Each edge was assigned: (i) mean confidence across all N replies between the user pair,

$$\bar{w} = \frac{1}{N} \sum_{i=1}^N w_i \quad (4)$$

and (ii) a confidence-weighted net stance vector $\mathbf{p} = [p_{\text{disagree}}, p_{\text{neutral}}, p_{\text{agree}}]$, defined as:

$$p_k = \frac{\sum_{i=1}^N \mathbf{1}(y_i = k) \cdot w_i}{\sum_{i=1}^N w_i}, \quad k \in \{\text{disagree}, \text{neutral}, \text{agree}\} \quad (5)$$

where y_i denotes the stance label of reply i . This ensures the vector components sum to one, preserving both stance polarity and annotation reliability.

5) *User Text Embeddings*: We extracted semantic representations of comments and submissions using the pre-trained Sentence-Transformers model `all-MiniLM-L6-v2`. Each comment embedding is a 384-dimensional vector, which serves as the basis for constructing user-level node features in the debate snapshot graphs (described in Sec. III-C).

TABLE I
PREPROCESSED DEBAGREEMENT DATASET STATISTICS. SUBREDDIT-LEVEL REPLY, STANCE, AND TEMPORAL STATISTICS (SELF-REPLIES REMOVED).

Subreddit	Earliest Date	Latest Date	# Replies	% Disagree	% Neutral	% Agree	# Unique Comments	# Unique Authors	Total TS	Avg TS Window (days)
All	2015-01-03	2021-05-19	42,770	39.6	26.0	34.4	75,034	27,420	77	69.3
blacklivesmatter	2020-05-27	2021-05-19	1,916	32.9	22.4	44.7	3,453	2,691	10	33.5
brexit	2016-06-21	2021-05-18	15,703	41.6	29.2	29.2	28,039	1,400	23	76.0
climate	2015-01-03	2021-05-18	5,754	40.6	27.7	31.7	10,131	5,377	26	87.0
democrats	2020-01-02	2021-05-19	9,606	35.9	21.9	42.2	16,499	8,187	9	55.0
republican	2020-01-02	2021-05-19	9,791	40.7	24.6	34.7	16,912	10,154	9	54.9

C. Graph Construction

We constructed separate directed graphs for each subreddit and temporal window. Nodes represent users, while edges denote aggregated reply interactions (as defined in Sec. III-B4).

1) *Node Features*: For each temporal window, we construct user-level semantic representations by mean-pooling the embeddings of all comments authored by each user. Users without embeddings in a given interval were excluded, since edges could only be formed between users with available pooled representations. Formally, for a user u at timestep t :

$$\mathbf{h}_{u,t} = \frac{1}{M} \sum_{j=1}^M \mathbf{z}_j, \quad (6)$$

where $\mathbf{z}_j$ denotes the embedding of comment j authored by u in timestep t , and M is the number of such comments. This approach follows Alatawi et al.’s implementation, which averages about 10 Twitter posts per user (Alatawi, Sheth, and Liu 2024). In our setting, we extend this to all available comments within a window, since user activity levels are highly uneven (some users contribute dozens of comments while others post once). By averaging over all contributions, we mitigate bias from prolific users while retaining the full semantic signal present in the data.

2) *Edge Features*: Each directed edge (u, v) consists of the feature vector: $\mathbf{e} = [\bar{w}, p_{\text{disagree}}, p_{\text{neutral}}, p_{\text{agree}}]$, where $\bar{w}$ is the mean confidence across all replies between the users (Eq. 4), and $\mathbf{p}$ is the confidence-weighted stance distribution (Eq. 5). Self-loops were excluded.

3) *Graph Filtering*: To reduce fragmentation, each graph snapshot is decomposed into weakly connected components (WCCs), retaining only those with at least 3 edges. This filtering removes isolated pairs, trivial interactions and singleton nodes, ensuring that community detection operates on substantive clusters. A WCC is defined as a maximal subgraph in which every pair of vertices is connected by some path when edge directions are ignored (Veerman and Kummel 2019), and may contain one or more strongly connected components (SCCs). We adopt WCCs rather than SCCs because reply graphs are inherently asymmetric; interactions typically flow one way (from a commenter to the replied-to user). Empirical studies of directed social networks also show that SCCs in such settings are often extremely small or trivial (Myers et al.

2014; Rodgers, Tiño, and Johnson 2023; Federico et al. 2024), so filtering at the SCC level risks over-fragmenting the graphs and discarding meaningful structure.

4) *Graph Formats*: To facilitate downstream analysis and training tasks, we store each graph snapshot in two formats: (i) a NetworkX DiGraph for community detection and structural analysis and (ii) a PyTorch Geometric Data object for model training, containing node features $X_t \in \mathbb{R}^{|V_t| \times d}$, a directed edge index matrix $\mathbf{E}_t^{\text{idx}} \in \mathbb{N}^{2 \times |E_t|}$ and edge attributes $E_t^f \in \mathbb{R}^{|E_t| \times f}$. A single graph snapshot is therefore formally defined as:

$$G_t = (V_t, E_t, X_t, \mathbf{E}_t^{\text{idx}}, E_t^f). \quad (7)$$

In total, 77 temporal graph objects were constructed across all subreddits. These graphs form the input to the GNN model and the subsequent community detection for ECS analysis.

D. Proposed Approach

1) *Message Passing in GNNs*: Message passing is the core mechanism by which modern GNNs propagate information over graphs. At layer l , each node i aggregates messages from its neighbours and updates its representation. Intuitively, the layer computes messages from neighbours (optionally using edge information), and then applies an update at the destination node. This iterative procedure lets node states encode increasingly wider relational context.

a) *Graph Convolutional Network (GCN)*: GCN (Kipf and Welling 2017) implements message passing through a localised spectral convolution of the graph Laplacian. In practice, this reduces to degree-normalised neighbourhood aggregation, followed by a shared linear transformation and non-linearity:

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}), \quad (8)$$

where self-loops are added via $\tilde{A} = A + I$, $\tilde{D}$ is the degree matrix and $W^{(l)}$ are learnable weights. In PyTorch Geometric, this operation is implemented by GCNConv. In its vanilla form, GCN aggregation does not incorporate edge attributes; all edges are treated uniformly apart from normalisation, which limits expressivity in debate graphs where edges encode stance and confidence.

b) *GraphSAGE*: GraphSAGE (Hamilton, Ying, and Leskovec 2017) introduces an inductive framework that learns a neighbourhood aggregation function applicable to unseen nodes. At each layer, it samples a fixed-size set of neighbours, aggregates their representations using a permutation-invariant operator (e.g., mean, LSTM, max pooling), and concatenates the result with the target node’s own features:

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot [h_i^{(l)} \parallel \text{AGG}^{(l)} \{h_j^{(l)} : j \in \mathcal{N}(i)\}] \right), \quad (9)$$

where σ is a non-linear activation, $\parallel$ denotes vector concatenation, and $\mathcal{N}(i)$ is the sampled neighbourhood of node i . In PyTorch Geometric, this is implemented by `SAGEConv`. As with GCNs, standard GraphSAGE does not condition aggregation on edge attributes. Edges define neighbourhood structure and sampling, but do not weight messages, which can be disadvantageous in debate graphs where stance proportions and confidence should modulate interaction strength.

c) *Edge-Conditioned Convolution (ECC)*: ECC (Simonovsky and Komodakis 2017) addresses the limitations of GCN and GraphSAGE by generating edge-specific filters from edge features. A filter network $F^{(l)}$ maps an edge feature vector $\mathbf{e}_{ji}$ (denoted $L(j, i)$ in (Simonovsky and Komodakis 2017)) to a weight matrix $\Theta_{ji}^{(l)} = F^{(l)}(\mathbf{e}_{ji})$, which modulates incoming messages. The node update rule is then:

$$X^{(l)}(i) = \frac{1}{|N(i)|} \sum_{j \in N(i)} \Theta_{ji}^{(l)} X^{(l-1)}(j) + b^{(l)}. \quad (10)$$

Figure 1 details our ECC implementation. Here, each directed edge (j, i) is represented by $\mathbf{e}_{(j,i)} = [\bar{w}_{(j,i)}, \mathbf{p}_{(j,i)}]$, where $\bar{w}_{(j,i)}$ denotes the mean annotator confidence and $\mathbf{p}_{(j,i)} = [p_{\text{disagree}}, p_{\text{neutral}}, p_{\text{agree}}]$ is the confidence-weighted ternary stance distribution. To enhance expressive capacity, the stance vector $\mathbf{p}_{(j,i)} \in \mathbb{R}^3$ is first mapped to a higher-dimensional embedding through a stance encoder $g : \mathbb{R}^3 \rightarrow \mathbb{R}^{16}$:

$$g(\mathbf{p}_{(j,i)}) = \phi(W_p \mathbf{p}_{(j,i)} + b_p), \quad (11)$$

where ϕ is a nonlinear activation. The concatenated representation $[\bar{w}_{(j,i)} \parallel g(\mathbf{p}_{(j,i)})] \in \mathbb{R}^{17}$ is then passed to the filter-generating network $F^{(l)}(\cdot)$, which produces the edge-specific transformation matrix:

$$\Theta_{ji}^{(l)} = F^{(l)}([\bar{w}_{(j,i)} \parallel g(\mathbf{p}_{(j,i)})]), \quad \Theta_{ji}^{(l)} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}. \quad (12)$$

To stabilise training, the generated edge filters are ℓ_2 -normalised, and `DropEdge` is applied to reduce oversmoothing. This ensures that messages are modulated not only by structural connectivity but also by stance and confidence, aligning with the argumentative dynamics of online debates.

2) Hybrid Convolution Framework:

a) *DebateGNN*: We propose **DebateGNN**, a hybrid GNN framework tailored for stance-aware debate graphs. The backbone comprises two `SAGEConv` layers followed by a final ECC layer, with LayerNorm, GELU activations and Dropout applied between layers. A skip projection maps the input features $X^0 \in \mathbb{R}^{384}$ to match the final embedding dimension,

yielding $X_{\text{skip}}^0 = W_{\text{skip}} X^0$, which is added to the ECC output X^9 before the GELU activation. We choose a three-layer configuration to capture multi-hop relational structure without over-smoothing, which is a known limitation of deeper GNNs (Q. Li, Han, and X.-M. Wu 2018; Oono and Suzuki 2020; Z. Wu et al. 2021). In contrast to the two-layer GCN encoder used in EchoGAE (Alatawi, Sheth, and Liu 2024), DebateGNN extends depth and hybridises convolution types to support richer relational encoding while maintaining stability. LayerNorm is employed instead of batch normalisation, as temporal subreddit graphs vary substantially in size and degree, making batch statistics highly unstable in GNNs due to excessive batch noise (Cai et al. 2021). Node-wise normalisation—equivalent to LayerNorm—operates on each node’s feature vector independently, without relying on batch or global graph statistics, and is therefore more robust in heterogeneous graph settings (Chen et al. 2022). Furthermore, we choose the Gaussian Error Linear Unit (GELU) activation, instead of a standard ReLU, as it has been shown to stabilise optimisation and improve representations in deep models like transformers and GNNs alike (Hendrycks and Gimpel 2016; Lee 2023).

A schematic of the DebateGNN architecture is shown in Figure 2, where the two GraphSAGE layers propagate inductive structure across temporally segmented subreddit graphs, while the final ECC layer injects stance and confidence information into the last message-passing step.

E. Model Pretraining

1) *Model Configuration*: Unless otherwise specified, our DebateGNN uses 384-dimensional input embeddings (generated in Sec. III-B5), projected to a 256-dimensional hidden layer and 128-dimensional output embeddings. Node dropout is set to 0.15, with an additional edge MLP dropout of 0.1 and a keep probability of 0.8. Edge attributes include a 1-dimensional confidence score and a 3-dimensional stance vector. Negative edge attributes are weighted by 0.3. Training tasks are equally weighted (1.0 each) unless otherwise stated.

2) *Training Protocol*: We adopt a leave-one-out cross-validation scheme at the subreddit level: in each fold, one subreddit is held out entirely as the test set. A second subreddit is randomly chosen as the validation set, where the most recent k timesteps are reserved for validation and earlier timesteps merged into training. All remaining subreddits are used for training, with graphs within each subreddit ordered temporally. To mitigate potential temporal leakage across the train-validation boundary, we also consider leaving a gap of g timesteps between the training and validation sets. Unless otherwise specified, we use $g = 1$.

Full details of the multitask training setup, evaluation metrics and optimisation configuration are provided in Appendix A.

F. Baseline for ECS Node Embedding

We adopt the EchoGAE framework of Alatawi et al. (Alatawi, Sheth, and Liu 2024) as both a baseline for DebateGNN comparison and a source of node embeddings for

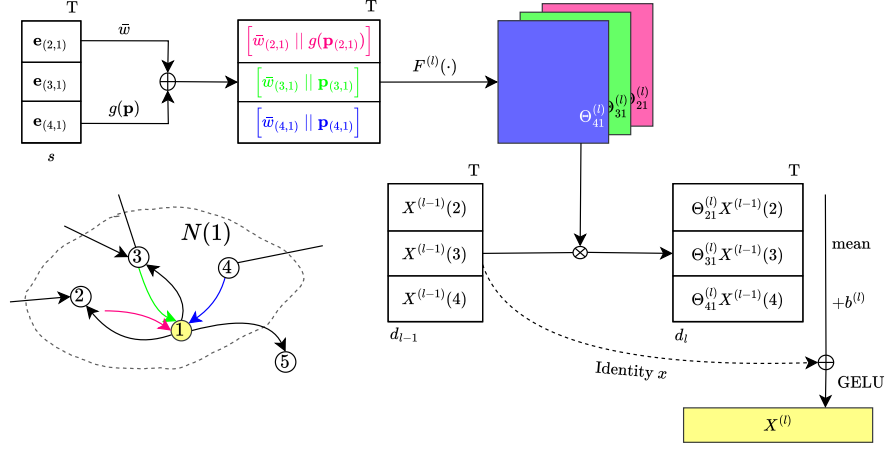


Fig. 1. **ECC message passing.** Each incoming edge (j, i) is represented by a feature vector $\mathbf{e}_{(j,i)} = [\bar{w}_{(j,i)} \parallel \mathbf{p}_{(j,i)}]$, where $\bar{w}$ is the mean confidence and $\mathbf{p}$ is the confidence-weighted stance distribution. The stance vector is first encoded by $g(\cdot)$ and concatenated with $\bar{w}$ before being passed to the filter-generating network $F^{(l)}(\cdot)$, which outputs edge-specific transformation matrices $\Theta_{ji}^{(l)}$. These matrices are applied to the neighbor features $X^{(l-1)}(j)$, aggregated over the neighborhood $\mathcal{N}(i)$, and combined with a root projection and bias. The result is passed through GELU to produce the updated node representation $X^{(l)}(i)$. Figure style inspired by Simonovsky and Komodakis 2017.

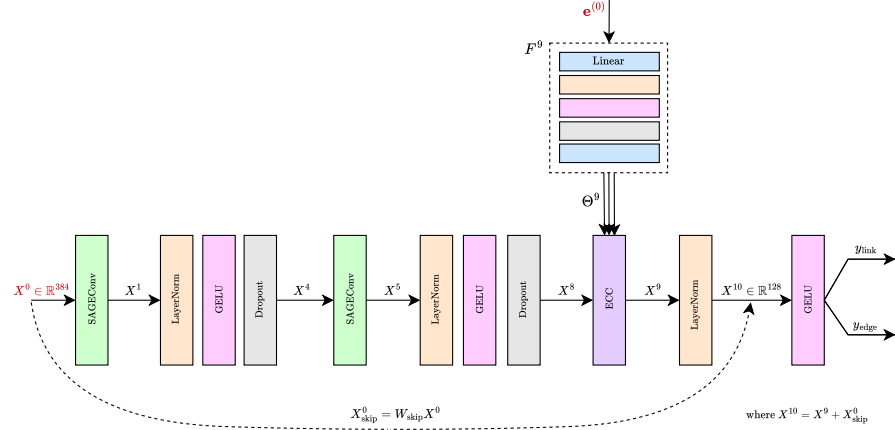


Fig. 2. **Multitask DebateGNN architecture.** Input node features $X^0 \in \mathbb{R}^{384}$ are processed by SAGE and ECC layers, with edge-conditioned filters generated from confidence and stance features. A skip projection $X_{skip}^0 = W_{skip} X^0$ ensures dimensional alignment, and the residual output $X^{10} = \text{GELU}(X^9 + X_{skip}^0)$ feeds into link and edge prediction heads. Figure style inspired by Simonovsky and Komodakis 2017.

Echo Chamber Score (ECS) computation. For consistency, we retain their default hyperparameters, including a fixed random seed of 42. EchoGAE employs a two-layer GCN encoder that projects inputs into a 100-dimensional hidden space and then to 50-dimensional output embeddings, with ReLU applied after the first layer. Training follows the original procedure, optimising a reconstruction loss over observed edges via positive and negative sampling.

IV. ECHO CHAMBER ANALYSIS

A. Community Detection

1) *Leiden Algorithm:* To generate community labels, we apply the Leiden community detection algorithm, which optimises modularity while accounting for the network’s degree sequence. As Leiden assumes undirected graphs, we

preserve directionality in our debate networks by converting them into stance-weighted undirected graphs, where each edge weight equals the agreement difference ($p_{\text{agree}} - p_{\text{disagree}}$) derived from the net stance vector $\mathbf{p}$ (Sec. III-B4). Although EchoGAE was evaluated on Louvain communities in the original study (Alatawi, Sheth, and Liu 2024), we adopt Leiden because it guarantees more connected communities and yields higher-quality partitions with faster performance, particularly on large or complex networks (Traag, Waltman, and Eck 2019). Both Louvain and Leiden share the same core idea behind modularity, i.e., maximising the difference between the actual number of edges observed inside a community and the expected number of edges under a null model (Traag, Waltman, and Eck 2019). Each snapshot graph is then in-

dividually optimised to select the resolution parameter that maximises modularity subject to three structural constraints: (i) at least 2 communities to avoid degenerate partitions, (ii) a minimum community size of 3 nodes to rule out trivial singletons and (iii) at most $0.8\sqrt{n}$ communities to prevent over-fragmentation, where n is the number of nodes in the graph.

B. Echo Chamber Metrics

For each subreddit snapshot graph, we apply the node embeddings learned by our DebateGNN and the baseline EchoGAE (Alatawi, Sheth, and Liu 2024) to downstream ECS analysis. We then compute echo chamber indices (ECIs) at both the community and overall levels, using the ECS Equations (1) and (2), for embeddings produced by both EchoGAE and DebateGNN.

C. ECS Community Analyser

After obtaining Leiden community labels and ECIs for both EchoGAE and DebateGNN, we track additional statistics per subreddit snapshot, including ECI deltas (Δ), modularity (Q) and silhouette scores (for measuring how well embeddings align with community structure). Community evolution is then assessed by computing Jaccard similarity, which quantifies the overlap in user membership between communities across timesteps. This allows us to identify retained, disappeared and emerged groups between network snapshots. From this taxonomy, we derive user retention, growth and migration rates, which are aggregated with ECS statistics and evaluated using paired t -tests, effect sizes and descriptive measures for temporal and network-level analyses.

V. RESULTS

A. Overall ECS Trends

Table II summarises EC statistics across five subreddits, while Figure 3A shows temporal evolution. Across all timesteps, EchoGAE yields a higher average ECI (0.454 ± 0.051) than DebateGNN (0.402 ± 0.081 ; $t = 6.611$, $p < 0.001$, $d = 0.758$). Silhouettes mirror this pattern: EchoGAE is less negative (-0.148 ± 0.101) than DebateGNN (-0.245 ± 0.159), indicating closer alignment with Leiden partitions.

In debate graphs, learning dominated by *undirected, attribute-agnostic* structure means that increases in within-group activity (including disagreement) can raise cohesion in the structural partition and thus *inflate* ECI when embeddings track topology. EchoGAE, which primarily aggregates over structural signals (even with text features), reflects this behaviour in its higher ECIs and silhouettes. DebateGNN, by contrast, incorporates *edge* stance and confidence into message passing, allowing cross-structural signals that dampen ECI but capture semantic dynamics not visible in topology alone.

Overall, EchoGAE produces higher ECI across all subreddits except **r/brexit**, which shows both the lowest modularity ($Q = 0.161 \pm 0.140$) and the highest share of disagreement (41.6%; Table I). Here, DebateGNN reports higher ECI (0.481 ± 0.037 vs. 0.462 ± 0.076) and a less-negative silhouette

(-0.067 ± 0.068 vs. -0.103 ± 0.147), suggesting a potential advantage for stance-aware modelling in low-modularity, disagreement-heavy settings.

B. Brexit Case Study

In r/brexit’s ECS evolution (Fig. 3B), EchoGAE and DebateGNN follow broadly similar ECI trajectories, but EchoGAE shows larger swings, while DebateGNN remains smoother and slightly higher on average. The purple Jaccard curve highlights intermittent troughs (e.g., T9–T10 and T13–T15), signalling community reshuffles; these coincide with short-lived dips in both ECIs, followed by recovery as the partition stabilises. The user-migration plot (Fig. 4) aligns with this: churn spikes (simultaneous rises in new and lost users with lower retention) cluster around the same transitions. Across these shocks, EchoGAE reacts more sharply, while DebateGNN is damped, suggesting steadier chamber estimates under rapid structural reconfiguration and shifting user mix. Taken together, continuity and migration evidence in r/brexit is *consistent with* DebateGNN providing steadier EC estimates under short-horizon shocks, whereas EchoGAE remains more sensitive to such shifts.

We examine snapshot **T12 (Aug–Oct 2019)** (Fig. 5), a heavy-traffic timestep in r/brexit (330 nodes, 1,414 edges; $Q = 0.277$; see per-timestep statistics in Appendix). T12 coincides with late-stage Brexit flashpoints: the Prime Minister’s *advice* to prorogue Parliament (late August), the UK Supreme Court’s unanimous ruling against prorogation (24 September), and the Government’s revised Irish backstop proposals (2 October) (Walker 2021; Cowie and Cygan 2020). EchoGAE and DebateGNN report comparable ECI values (0.464 vs. 0.472). In both UMAPs, Leiden communities appear interleaved, consistent with modestly negative silhouettes; DebateGNN’s silhouette is marginally less negative (-0.073 vs. -0.088). Because UMAP is a non-linear projection, we interpret these qualitatively: EchoGAE forms a long arc with gradual colour mixing, whereas DebateGNN shows denser local pockets and a small compact island—though mixing remains evident in both.

Takeaway. In a high-volume window within r/brexit—a low-modularity and disagreement-heavy subreddit—both methods yield comparable ECI at T12, and neither cleanly separates Leiden communities. DebateGNN is marginally closer to the partition (slightly less negative silhouette). Across the broader r/brexit case study, churn periods coincide with sharper EchoGAE swings and comparatively steadier DebateGNN estimates. Practically, topology-dominated embeddings (even with text features) may overstate *apparent* EC intensity during high-activity windows, whereas incorporating edge stance and confidence produces more conservative, churn-resilient estimates.

VI. DISCUSSION

Implications. Because silhouettes are computed on embeddings w.r.t. Leiden labels, they quantify *alignment to the structural partition* rather than inherent ideological separation.

TABLE II

SUBREDDIT-LEVEL ECS SUMMARY (MEAN $\pm$ SD ACROSS TIMESTEPS). COLUMNS: # TIMESTEPS, MODULARITY Q , MEAN ECI-ECHO GAE (ALATAWI, SHETH, AND LIU 2024) AND DEBATEGNN (OURS), Δ ECI (*drift = mean signed per-timestep change; can be negative*) AND SILHOUETTE. **ALL** AGGREGATES ALL SUBREDDITS. ECHO GAE GENERALLY SHOWS HIGHER ECIS AND LESS-NEGATIVE SILHOUETTES; **R/BREXIT** IS THE EXCEPTION, WHERE DEBATEGNN IS SLIGHTLY HIGHER AND LESS NEGATIVE.

Subreddits	# Timesteps	Modularity (Q)	EchoGAE ECI (Alatawi, Sheth, and Liu 2024)	DebateGNN ECI (Ours)	EchoGAE Δ ECI (Alatawi, Sheth, and Liu 2024)	DebateGNN Δ ECI (ours)	EchoGAE Silhouette (Alatawi, Sheth, and Liu 2024)	DebateGNN Silhouette (Ours)
ALL	77	0.365 ± 0.279	0.454 ± 0.051	0.402 ± 0.081	0.001 ± 0.057	0.001 ± 0.052	-0.148 ± 0.101	-0.245 ± 0.159
blacklivesmatter	10	0.852 ± 0.069	0.472 ± 0.014	0.403 ± 0.070	0.006 ± 0.016	0.010 ± 0.061	-0.212 ± 0.038	-0.324 ± 0.104
brexit	23	0.161 ± 0.140	0.462 ± 0.076	0.481 ± 0.037	0.002 ± 0.094	-0.002 ± 0.043	-0.103 ± 0.147	-0.067 ± 0.068
climate	26	0.431 ± 0.242	0.466 ± 0.032	0.399 ± 0.053	-0.001 ± 0.032	0.000 ± 0.061	-0.129 ± 0.068	-0.254 ± 0.105
democrats	9	0.180 ± 0.109	0.414 ± 0.025	0.307 ± 0.040	0.004 ± 0.040	0.006 ± 0.056	-0.197 ± 0.048	-0.404 ± 0.078
republican	9	0.342 ± 0.122	0.421 ± 0.021	0.299 ± 0.046	-0.001 ± 0.033	0.000 ± 0.044	-0.198 ± 0.038	-0.431 ± 0.089

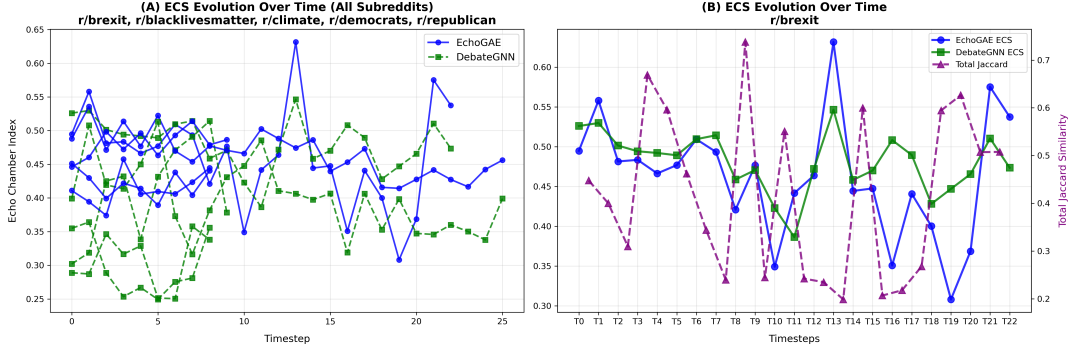


Fig. 3. **ECS evolution and community continuity.** (A) ECI trajectories across five subreddits for EchoGAE (blue) and DebateGNN (green); the methods differ in both level and volatility. (B) r/brexit detail with Total Jaccard (purple, right axis) as a community-continuity proxy.

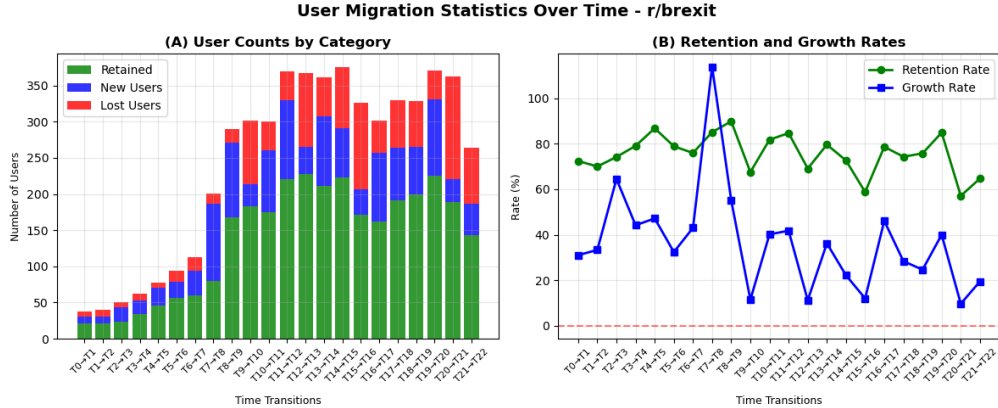


Fig. 4. **ECS evolution and community continuity.** (A) ECI trajectories across five subreddits for EchoGAE (blue) and DebateGNN (green); the methods differ in both level and volatility (see also Tab. II for subreddit-level aggregates). (B) r/brexit detail with Total Jaccard (purple, right axis) as a community-continuity proxy. Communities via Leiden (RB) on stance-weighted graphs; edge weight = agreement difference ($p_{agree} - p_{disagree}$) derived from the net stance vector $\mathbf{p}$ (Sec. IV-A1).

In debate graphs, undirected and attribute-agnostic interactions (both agreement and disagreement) can increase within-cluster cohesion as activity rises. When the encoder closely mirrors topology, the *measured* ECI can therefore be inflated. In our data, EchoGAE appears more structure-led and reactive to rapid reconfigurations, whereas DebateGNN’s incorporation of edge stance and confidence signals often dampens ECIs and yields steadier estimates under churn. The two models thus emphasise *topology-dominant sensitivity* versus *stance-aware stability*. More broadly, both silhouette and ECI are partition-dependent; these metrics should be interpreted as contingent

on graph structure, edge-weighting, time window, community definition (e.g., Leiden and its resolution) and metric design, rather than as absolute indicators.

Limitations.

- **Partition dependence and resolution sensitivity.** Both ECI and silhouette are computed w.r.t. Leiden and its chosen resolution; WCC filtering can exclude small or minority clusters.
- **Temporal uncertainty and binning.** DEBAGREEMENT consists of depth-1 pairs with incomplete timestamps; fixed windowing (with merges) introduces sensi-

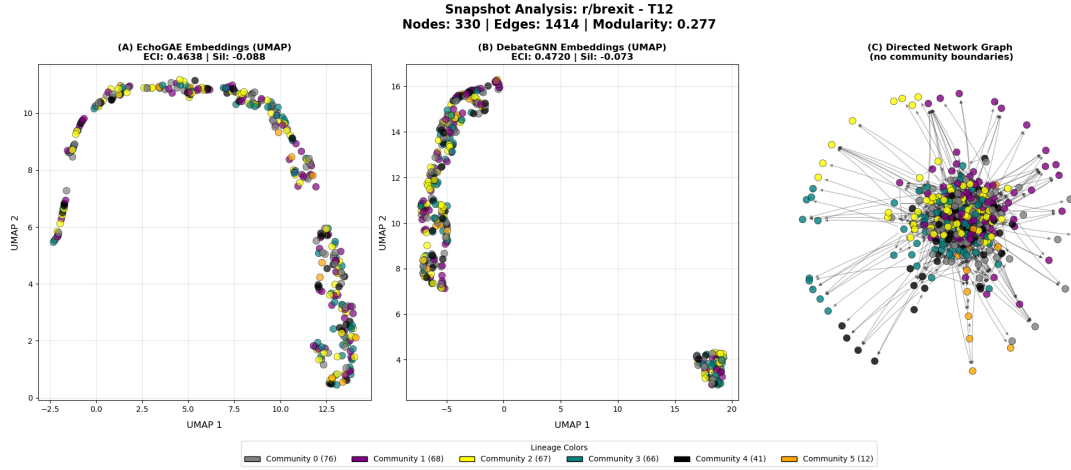


Fig. 5. Snapshot analysis of r/brexit at T12 at T12 (2019-08-05 – 2019-10-03). (A) EchoGAE and (B) DebateGNN 2D UMAPs of node embeddings, coloured by Leiden (RB) communities. Both projections show substantial interleaving of Leiden community colours, consistent with modest (slightly negative) silhouettes; DebateGNN forms denser local pockets and a small compact island (ECI: 0.472 vs. 0.464; Sil.: -0.073 vs. -0.088). (C) Directed interaction graph for the same timestep shows a dense, intermixed structure characteristic of high-activity debate periods.

tivity to bin size and segmentation.

- **Conversation flattening.** Averaging multiple replies within a dyad removes sequential order and escalation effects, potentially masking stance shifts within the same dyad.
- **Architecture scope (no explicit dynamics).** In 2-SAGE-1-ECC, stance and confidence modulate only in the final layer; the model is static and may under-capture finer temporal dependencies.
- **Statistical dependence.** Pooled paired tests treat subreddit-timestep units as approximately independent, but temporal autocorrelation may bias significance estimates.

Future Work.

- Report both topology-led and stance-aware ECI series, and optionally include a descriptive difference Δ_t to contextualise episodes (conditioned on the chosen partition and parameters).
- Complement ECI with simple auxiliary signals such as activity volume, churn, and partition continuity (e.g., Jaccard similarity across windows).
- Incorporate minimal sensitivity checks (e.g., alternative partitioners or edge semantics) and provide uncertainty summaries (e.g., confidence intervals) where feasible.

Ethical Considerations. Because EC findings are sensitive to operationalisation and often volatile, they should be interpreted with caution. Future work should avoid individual-level inferences and ensure that any downstream use (e.g., ranking or intervention) includes human oversight and opt-in transparency.

VII. ABLATION STUDY

To assess the contribution of individual design choices in DebateGNN, we conduct a series of ablation experiments (Appendix B). We adopt the proxy-task assumption that higher link AUC and lower confidence/stance NLL (and stance KL)

indicate more informative embeddings, while recognising that ECS is contingent on the chosen community partition and is therefore reported separately at the system level.

VIII. CONCLUSION

This thesis addressed key limitations in echo chamber (EC) research, which has often relied on homophily or endorsement proxies while overlooking the stance-rich and temporal nature of debates. We introduced DebateGNN, a stance-aware hybrid GNN, and extended the Echo Chamber Score (ECS) to temporal Reddit reply graphs. Our results show that conditioning message passing on edge stance and annotator confidence alters ECS behaviour (i.e., yielding lower average EC intensity but higher variance across timesteps), reflecting greater sensitivity to disagreement signals. EchoGAE generally reports higher mean ECI and less-negative silhouettes, indicating embeddings more closely aligned with topology-derived Leiden partitions. By contrast, DebateGNN estimates diverge more from topology; in disagreement-heavy, low-modularity settings such as r/brexit, DebateGNN can exceed EchoGAE while exhibiting steadier trajectories during churn episodes.

The key methodological contribution is that ECS outcomes are contingent on modelling choices—including partitioning and edge semantics—and should therefore be treated as contextual diagnostics rather than absolute measures. Practically, presenting both topology-led and stance-aware estimates offers a more balanced view. Future work should incorporate explicit temporal dynamics and evaluate robustness across datasets and partitioning methods.

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APPENDIX A DEBATEGNN TRAINING SETUP AND EVALUATION METRICS

This appendix provides the full training setup and evaluation details omitted from the main text for brevity. We describe the multitask learning formulation of DebateGNN (loss functions and weighting) followed by the metrics used to assess performance across link, confidence and stance prediction tasks.

A. Multitask Learning Setup

Given candidate node embeddings $[z_j \| z_i]$, we jointly optimise three edge-level objectives using two prediction heads: a link head for edge existence and a shared edge head for confidence and stance.

1) *Link Prediction*: The link head produces a logit $\hat{s}_{ij}$. Training uses observed edges as positives and sampled non-edges as negatives. We minimise binary cross-entropy with logits using PyTorch’s `binary_cross_entropy_with_logits`:

$$\mathcal{L}_{\text{link}} = \text{BCEWithLogits}(\hat{s}_{ij}, y_{ij}, w^+), \quad (\text{A.1})$$

where $y_{ij} \in \{0, 1\}$, and w^+ is a positive class weight.

2) *Confidence Prediction*: The edge head outputs a scalar $\hat{c}_{ij}$ that is trained against the mean confidence $\bar{w}_{ij}$. For negatives, we assign a softened target near zero rather than a hard zero, then apply per-sample weights to prevent imbalance. The loss is a Smooth L1 objective implemented with `torch.nn.SmoothL1Loss`:

$$\mathcal{L}_{\text{conf}} = \text{SmoothL1}(\hat{c}_{ij}, \bar{w}_{ij}). \quad (\text{A.2})$$

3) *Stance Prediction*: The same edge head also outputs a distribution $\hat{\mathbf{p}}_{ij}$ over $\{\text{disagree}, \text{neutral}, \text{agree}\}$, trained against soft targets $\mathbf{p}_{ij}$. For positive edges, $\mathbf{p}_{ij}$ corresponds to annotator-derived net stance distributions; for negatives, to avoid collapse, we assign a class weight beforehand. Using `torch.nn.functional.kl_div`, we minimise the KL divergence:

$$\mathcal{L}_{\text{stance}} = \text{KLDiv}(\hat{\mathbf{p}}_{ij}, \mathbf{p}_{ij}). \quad (\text{A.3})$$

4) *Uncertainty-Weighted Multitask Objective*: Following Cipolla, Gal, and Kendall (2018), each task loss $\mathcal{L}_j$ is weighted by an uncertainty parameter σ_j :

$$\mathcal{L}_j^* = \frac{1}{2\sigma_j^2} \mathcal{L}_j + \log \sigma_j. \quad (\text{A.4})$$

TABLE B.1
BENCHMARK COMPARISON. LINK PREDICTION AUC FOR DEBATEGNN BENCHMARK (2-SAGE-1-ECC) AND HOMOGENEOUS GCN/SAGE BASELINES.

	2-SAGE-1-ECC	3-GCN	3-SAGE
Link AUC	0.875	0.750	0.866

In practice, this is reparameterised by a learnable log-variance $s_j = \log \sigma_j^2$, yielding the more stable form:

$$\mathcal{L}_j^* = \frac{1}{2} e^{-s_j} \mathcal{L}_j + \frac{1}{2} s_j. \quad (\text{A.5})$$

The total objective is then the sum across active tasks:

$$\mathcal{L}_{\text{total}} = \sum_{j \in \{\text{link}, \text{conf}, \text{stance}\}} \mathcal{L}_j^*. \quad (\text{A.6})$$

B. Evaluation Metrics

We report task-specific performance using the following metrics:

- **Link prediction**: Area Under the ROC Curve (AUC), measuring the discriminative ability of the link head.
- **Confidence prediction**: Mean Absolute Error (MAE) for calibration accuracy, and Negative Log-Likelihood (NLL) for probabilistic fit.
- **Stance prediction**: KL divergence between predicted and target stance distributions, and Negative Log-Likelihood (NLL) for categorical likelihood.

C. Training Configuration

Models are trained for up to 30 epochs on batches of individual subreddit graphs, using the Adam optimiser with decoupled learning rates for base parameters (1×10^{-4}) and log-variance parameters (1×10^{-5}). Weight decay is set to 1×10^{-4} , with a cosine annealing scheduler and warm restarts ($T_0 = 20$). Early stopping is triggered after 6 epochs without a validation improvement greater than 3×10^{-4} . Negative sampling is applied at a 0.5 ratio relative to observed edges, and temporal regularisation is applied with a weight of 0.1.

APPENDIX B DEBATEGNN ABLATION STUDY

We report the full results of ablation experiments described in Section VII. We expand on benchmark comparisons, layer composition variants, and validation gap sensitivity, providing complete tables and numerical values for reference.

A. Benchmark Comparison

We compare DebateGNN against homogeneous GNN backbones (SAGE and GCN) to evaluate the benefit of hybridisation. Since these baselines do not incorporate edge attributes, evaluation is restricted to link prediction performance (AUC). Results in Table B.1 show that the hybrid DebateGNN (2-SAGE-1-ECC) achieves the highest link prediction score (AUC = 0.875), outperforming both 3-SAGE and 3-GCN.

TABLE B.II
LAYER COMPOSITION COMPARISON. PERFORMANCE METRICS OF DEBATEGNN ACROSS DIFFERENT SAGE–ECC VARIANTS.

Model Variant	Link AUC	Confidence MAE	Confidence NLL	Stance NLL	Stance KL
2–SAGE–1–ECC (Baseline)	0.886	0.342	0.710	0.593	0.585
1–SAGE–2–ECC	0.815	0.335	0.734	0.600	0.592
1–ECC–2–SAGE	0.895	0.351	0.713	1.010	1.002
3–ECC	0.849	0.335	0.725	0.612	0.604

TABLE B.III
VALIDATION GAP ABLATION. MEAN TEST PERFORMANCE ACROSS FOLDS. A ONE-STEP GAP ($g = 1$) ACHIEVES THE BEST OVERALL PERFORMANCE.

Gap Size (g)	Link AUC	Confidence MAE	Confidence NLL	Stance NLL	Stance KL
1 (Baseline)	0.879	0.344	0.707	0.605	0.597
0	0.878	0.344	0.718	0.611	0.602
2	0.868	0.339	0.708	0.606	0.598
3	0.870	0.343	0.720	0.656	0.648

B. Layer Composition Variants

Table B-B summarises the effect of varying SAGE and ECC ordering. The 2-SAGE–1-ECC stack provides the best overall trade-off, combining high link AUC (0.886) with the lowest stance losses (NLL = 0.593; KL = 0.585). By contrast, 1-ECC–2-SAGE achieves the highest link AUC (0.895) but weaker stance calibration, while 1-SAGE–2-ECC underperforms on link prediction. The homogeneous 3-ECC stack is competitive on confidence tasks but inferior on stance prediction. Overall, hybridisation with two SAGE layers followed by ECC is most effective.

C. Validation Gap Sensitivity

Results in Table B-C show that performance is stable across gaps. Introducing a one-step gap ($g = 1$) slightly improves stance calibration (NLL = 0.605; KL = 0.597) without affecting link prediction (AUC = 0.879). Larger gaps ($g = 2, 3$) yield marginal declines, suggesting DebateGNN is robust to boundary leakage and that $g = 1$ provides a balanced setting.

APPENDIX C

ECS STATISTICS FOR R/BREXIT ACROSS ALL TIMESTEPS

Table C.IV reports the complete timestep-level statistics for the r/brexit case study. For each interval, we list graph size, modularity, detected community counts and ECI values obtained from both EchoGAE and DebateGNN. This table complements the main text by providing the full numeric results for the case study analysis.

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TABLE C.IV
COMPLETE ECS STATISTICS FOR R/BREXIT ACROSS ALL TIMESTEPS. EACH ROW CORRESPONDS TO ONE TEMPORAL INTERVAL WITH ITS ASSOCIATED GRAPH STATISTICS (NODE/EDGE COUNTS, MODULARITY Q AND NUMBER OF DETECTED COMMUNITIES), ALONG WITH ECIS COMPUTED USING ECHOgAE (ALATAWI, SHETH, AND LIU 2024) AND DEBATEGNN (OURS) EMBEDDINGS.

Timestep	Interval	# Nodes	# Edges	Modularity (Q)	# Communities	EchoGAE ECI (Alatawi, Sheth, and Liu 2024)	DebateGNN ECI (Ours)
0	2016-06-21 – 2017-10-13	29	84	0.238	3	0.492	0.526
1	2017-10-14 – 2017-12-12	30	79	0.213	2	0.558	0.530
2	2017-12-13 – 2018-02-10	31	59	0.064	2	0.481	0.501
3	2018-02-11 – 2018-04-11	43	107	0.305	4	0.483	0.494
4	2018-04-12 – 2018-06-10	53	134	0.328	4	0.466	0.492
5	2018-06-11 – 2018-08-09	71	266	0.279	3	0.477	0.489
6	2018-08-10 – 2018-10-08	79	187	0.296	2	0.509	0.509
7	2018-10-09 – 2018-12-07	94	176	0.033	2	0.493	0.514
8	2018-12-08 – 2019-02-05	187	611	0.339	6	0.421	0.459
9	2019-02-06 – 2019-04-06	271	1090	0.303	6	0.477	0.471
10	2019-04-07 – 2019-06-05	214	555	0.014	3	0.349	0.423
11	2019-06-06 – 2019-08-04	261	943	0.017	3	0.442	0.386
12	2019-08-05 – 2019-10-03	330	1414	0.277	6	0.464	0.472
13	2019-10-04 – 2019-12-02	265	697	0.009	2	0.632	0.546
14	2019-12-03 – 2020-01-31	307	1196	0.300	7	0.444	0.458
15	2020-02-01 – 2020-03-31	291	1194	0.296	6	0.448	0.470
16	2020-04-01 – 2020-05-30	206	606	0.013	2	0.351	0.508
17	2020-05-31 – 2020-07-29	257	1007	0.295	5	0.441	0.490
18	2020-07-30 – 2020-09-27	264	947	0.004	2	0.400	0.428
19	2020-09-28 – 2020-11-26	265	892	0.004	2	0.308	0.447
20	2020-11-27 – 2021-01-25	331	1377	0.025	3	0.369	0.466
21	2021-01-26 – 2021-03-26	221	542	0.022	2	0.575	0.510
22	2021-03-27 – 2021-05-18	186	417	0.028	3	0.537	0.474

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